

Definition 1 (Abstractor). An *abstractor* of valence $s_1, \dots, s_k.s$ is a phrase

$$x_1, \dots, x_k.a,$$

abbreviated $\vec{x}.a$, where $x_1, \dots, x_k$ are distinct variables with x_i of sort s_i , and a is an abt of sort s in which each x_i is bound. For $k = 0$ it is just a .

How to write it.

source	output
$\mathbf{x.a}$	$x.a$
$\backslash \mathbf{vec}\{\mathbf{x}\}.\mathbf{a}$	$\vec{x}.a$
$\mathbf{x_1}, \backslash \mathbf{dots}, \mathbf{x_k.a}$	$x_1, \dots, x_k.a$
$\backslash \mathbf{Exp}.\backslash \mathbf{Exp}$	$\mathbf{Exp.Exp}$
$(\backslash \mathbf{Exp}, \backslash \mathbf{Exp}.\backslash \mathbf{Exp})\backslash \mathbf{Exp}$	$(\mathbf{Exp}, \mathbf{Exp.Exp})\mathbf{Exp}$
$\backslash \mathbf{op}\{\mathbf{let}\}(\mathbf{a_1} \backslash, ; \backslash, \mathbf{x.a_2})$	$\mathbf{let}(a_1 ; x.a_2)$
$\backslash \mathbf{op}\{\mathbf{lam}\}[\backslash \mathbf{tau}](\mathbf{x.a})$	$\mathbf{lam}[\tau](x.a)$
$\mathbf{[b/x]a}$	$[b/x]a$
$\mathbf{a =_}\backslash \mathbf{alpha} \mathbf{b}$	$a =_\alpha b$

Formation rule (for reference).

$$\frac{\mathcal{X}, \vec{x}_1 \vdash a_1 : s_1 \quad \dots \quad \mathcal{X}, \vec{x}_n \vdash a_n : s_n}{\mathcal{X} \vdash o(\vec{x}_1.a_1 ; \dots ; \vec{x}_n.a_n) : s} \text{ OP}$$

where o has arity $(\vec{s}_1.s_1, \dots, \vec{s}_n.s_n)s$.